

8m * Matrix of $L.T [T]_{\beta}$

8m * Eigen values & Eigen vectors.

8m (or) 16m * Eigen values & Eigen vectors - Ordered Basis.

* - Diagonal matrix.

16m * Diagonalization.

UNIT V Inner Product Space:

Definition:

Let V be a vector space over $\mathbb{F}$. And inner product on V is a function that assigns to every ordered pair of vectors x and y in V , a scalar in $\mathbb{F}$, denoted by $\langle x, y \rangle$, such that for all $x, y, z \in V$ & $\alpha \in \mathbb{F}$, such that the following properties are holds good.

$$i) \langle x+y, z \rangle = \langle x, z \rangle + \langle y, z \rangle$$

$$ii) \langle \alpha x, y \rangle = \alpha \langle x, y \rangle$$

$$iii) \langle x, y \rangle = \overline{\langle y, x \rangle} \text{ where } \langle x, y \rangle \text{ is a complex conjugate}$$

$$iv) \langle x, x \rangle \geq 0 \text{ if } x \neq 0$$

1) show that $F^n(C)$ is an inner product space with respect to the inner product

defined as $\langle x, y \rangle = \sum_{i=1}^n a_i \bar{b}_i$

where $x = (a_1, a_2, \dots, a_n)$ &

$y = (b_1, b_2, \dots, b_n)$.

To show that : $F^n(C)$ is an inner product space w.r. to inner product

Given:

$$\langle x, y \rangle = \sum_{i=1}^n a_i \bar{b}_i$$

where $x = (a_1, a_2, \dots, a_n) \in F^n(C)$

$y = (b_1, b_2, \dots, b_n) \in F^n(C)$

Let $z = (c_1, c_2, \dots, c_n) \in F^n(C)$

i) L.H.S:

$$\langle x+y, z \rangle = \langle (a_1+b_1, a_2+b_2, \dots, a_n+b_n), (c_1, c_2, \dots, c_n) \rangle$$

$$= \sum_{i=1}^n (a_i + b_i) \bar{c}_i$$

$$= \sum_{i=1}^n a_i \bar{c}_i + \sum_{i=1}^n b_i \bar{c}_i \quad (ii)$$

$$= \langle x, z \rangle + \langle y, z \rangle \quad \text{R.H.S}$$

ii) L.H.S:

$$\langle cx, y \rangle = \sum_{i=1}^n c a_i \bar{b}_i$$

$$= c \sum_{i=1}^n a_i \bar{b}_i$$

$$= c \langle x, y \rangle \text{ // } \underline{\underline{R.H.S}}$$

iii) R.H.S:

$$\langle y, x \rangle = \sum_{i=1}^n b_i \bar{a}_i \longrightarrow \textcircled{1}$$

L.H.S:

$$\overline{\langle x, y \rangle} = \overline{\left[\sum_{i=1}^n a_i \bar{b}_i \right]}$$

$$= \overline{a_1 \bar{b}_1 + a_2 \bar{b}_2 + \dots + a_n \bar{b}_n}$$

$$= \overline{a_1} \overline{\bar{b}_1} + \overline{a_2} \overline{\bar{b}_2} + \dots + \overline{a_n} \overline{\bar{b}_n}$$

$$= \overline{a_1} b_1 + \overline{a_2} b_2 + \dots + \overline{a_n} b_n$$

$$[\because \overline{\overline{x}} = x]$$

$$= b_1 \overline{a_1} + b_2 \overline{a_2} + \dots + b_n \overline{a_n}$$

$$= \sum_{i=1}^n b_i \bar{a}_i \longrightarrow \textcircled{2}$$

From $\textcircled{1} \propto \textcircled{2}$, we get

$$L.H.S = R.H.S.$$

$$\therefore \langle \overline{x}, y \rangle = \langle y, x \rangle //$$

$$\text{iv) } \langle x, x \rangle = \sum_{i=1}^n a_i \bar{a}_i$$

$$= \sum_{i=1}^n |a_i|^2$$

$$\text{since } |a_i|^2 \geq 0$$

$\therefore \langle x, x \rangle > 0$ if $x \neq 0$. Hence $F^n(\mathbb{C})$ is an inner product space with respect to

inner product.

$\therefore$ Hence proved.

1) ii) Show that $\mathbb{F}^n(\mathbb{R})$ is an inner product space with respect to the inner product defined as $\langle x, y \rangle = \sum_{i=1}^n a_i b_i$

where $x = (a_1, a_2, \dots, a_n)$ &

$y = (b_1, b_2, \dots, b_n)$.

To show that : $\mathbb{F}^n(\mathbb{R})$ is an inner product space w.r. to inner product.

Given :

$$\langle x, y \rangle = \sum_{i=1}^n a_i b_i$$

where $x = (a_1, a_2, \dots, a_n) \in \mathbb{F}^n(\mathbb{R})$

$y = (b_1, b_2, \dots, b_n) \in \mathbb{F}^n(\mathbb{R})$

Let $z = (c_1, c_2, \dots, c_n) \in \mathbb{F}^n(\mathbb{R})$

1) L.H.S :

$$\langle x+y, z \rangle = \langle (a_1+b_1, a_2+b_2, \dots, a_n+b_n),$$

$$(c_1, c_2, \dots, c_n) \rangle$$

$$= \sum_{i=1}^n (a_i+b_i) c_i$$

$$= \sum_{i=1}^n a_i c_i + \sum_{i=1}^n b_i c_i$$

$$= \langle x, z \rangle + \langle y, z \rangle$$

ii) L.H.S:

$$\langle x, y \rangle = \sum_{i=1}^n a_i b_i$$

$$= c \sum_{i=1}^n a_i b_i$$

$$= c \langle x, y \rangle$$

iii) R.H.S:

$$\langle y, x \rangle = \sum_{i=1}^n b_i a_i \rightarrow \textcircled{1}$$

L.H.S:

$$\langle \overline{x}, y \rangle = \sum_{i=1}^n \overline{a_i b_i} = \overline{b_1 a_1 + \dots + b_n a_n}$$

$$= \left(\sum_{i=1}^n \overline{a_i} \overline{b_i} \right) \quad a_i, b_i \in \mathbb{R}, b_i \in \mathbb{R}$$

$$= \sum_{i=1}^n \overline{a_i} \overline{b_i} \quad [\because \overline{a_i} = a_i, \overline{b_i} = b_i]$$

$$= \sum_{i=1}^n b_i a_i \rightarrow \textcircled{2}$$

From $\textcircled{1}$ & $\textcircled{2}$, we get

$$\langle \overline{x}, y \rangle = \langle y, x \rangle$$

$$\text{iv) } \langle x, x \rangle = \sum_{i=1}^n a_i a_i$$

$$= \sum_{i=1}^n a_i^2$$

since $a_i^2 > 0$

$\therefore \langle x, x \rangle \geq 0$ when $x \neq 0$

Thus $F^n(\mathbb{R})$ is an inner product space with respect to inner product.

Hence proved.

1) iii) A/A/25 Prove that $\langle f, g \rangle = \int_0^1 f(t)g(t) dt$ is an inner product where $f, g \in \mathbb{R}[0,1]$ a vector space of real valued continuous function on $[0,1]$.

Solution:

To show that: $V \in \mathbb{R}[0,1]$ is an inner product space with respect to inner product.

Given:

$$\begin{aligned} \langle f, g \rangle &= \int_0^1 f(t)g(t) dt \\ &= \int_0^1 [f(t) + g(t)]h(t) dt \\ &= \int_0^1 f(t)h(t) dt + \int_0^1 g(t)h(t) dt \\ &= \langle f, h \rangle + \langle g, h \rangle \\ &= \text{RHS} \end{aligned}$$

ii) L.H.S:

$$\begin{aligned} \langle cf, g \rangle &= \int_0^1 cf(t)g(t) dt \\ &= c \int_0^1 f(t)g(t) dt \\ &= c \langle f, g \rangle \\ &= \text{R.H.S} \end{aligned}$$

iii) R.H.S:

$$\langle g, f \rangle = \int_0^1 g(t) f(t) dt \rightarrow \textcircled{1}$$

L.H.S:

$$\langle \overline{f}, g \rangle = \left[\int_0^1 \overline{f(t)} g(t) dt \right]$$

$$= \int_0^1 \overline{f(t)} \cdot \overline{g(t)} dt \quad (\text{By property of conjugate})$$

$$= \int_0^1 f(t) \cdot g(t) dt \quad [\because \overline{f(t)} = f(t) \\ \overline{g(t)} = g(t)]$$

$$= \int_0^1 g(t) \cdot f(t) dt \rightarrow \textcircled{2}$$

From $\textcircled{1}$ & $\textcircled{2}$, we get

$$\langle \overline{f}, g \rangle = \langle g, f \rangle$$

$$\text{iv) } \langle f, f \rangle = \int_0^1 f(t) \cdot f(t) dt$$

$$= \int_0^1 (f(t))^2 dt$$

Since $(f(t)) \geq 0$ which is non-negativity

$$\therefore \langle f, f \rangle \geq 0 \text{ when } f \neq 0$$

Thus, $V = \mathbb{R}[0, 1]$ is an inner product space with respect to inner product.

Hence proved.

1) iv) Prove that $\langle f, g \rangle = \int_0^1 f(t) \overline{g(t)} dt$ is an inner product where $f, g \in C[0,1]$ a vector space of complex valued continuous functions on $[0,1]$.

Solution:

To show that: $V = C[0,1]$ is an inner product space with respect to inner product.

Given:

$$\langle f, g \rangle = \int_0^1 f(t) \cdot \overline{g(t)} dt$$

$$\text{Let } h \in C[0,1]$$

i) L.H.S:

$$\langle f+h, g \rangle = \int_0^1 (f+h)(t) \overline{g(t)} dt$$

$$= \int_0^1 [f(t) + h(t)] \overline{g(t)} dt$$

$$= \int_0^1 f(t) \overline{g(t)} dt + \int_0^1 h(t) \overline{g(t)} dt$$

$$= \langle f, g \rangle + \langle h, g \rangle$$

ii) L.H.S:

$$\langle cf, g \rangle = \int_0^1 cf(t) \overline{g(t)} dt$$

$$= c \int_0^1 f(t) \overline{g(t)} dt$$

$$= c \langle f, g \rangle$$

$$= R.H.S.$$

iii) R.H.S:

$$\langle g, f \rangle = \int_0^1 g(t) \overline{f(t)} dt \rightarrow \textcircled{1}$$

L.H.S:

$$\langle \overline{f}, g \rangle = \left[\int_0^1 \overline{f(t)} g(t) dt \right]$$

$$= \int_0^1 \overline{f(t)} \cdot \overline{\overline{g(t)}} dt \quad (\text{By property of conjugate})$$

$$= \int_0^1 \overline{f(t)} \cdot g(t) dt \quad [\because \overline{\overline{z}} = z]$$

$$= \int_0^1 g(t) \cdot \overline{f(t)} dt \rightarrow \textcircled{2}$$

From $\textcircled{1}$ & $\textcircled{2}$, we get

$$L.H.S = R.H.S$$

$$\langle \overline{f}, g \rangle = \langle g, f \rangle$$

$$\text{iv) } \langle f, f \rangle = \int_0^1 f(t) \cdot \overline{f(t)} dt$$

$$= \int_0^1 |f(t)|^2 dt$$

$$\text{since } |f(t)| \geq 0$$

$$\therefore \langle f, f \rangle > 0 \text{ when } f \neq 0$$

$\therefore$ Thus, $V = C[0,1]$ is an inner product space with respect to inner product.

Hence proved.

Orthogonality:

Let x and y be vectors in an inner product space V . Then x is said to be orthogonal to y , if

$$\langle x, y \rangle = 0$$

Orthogonal Set:

A subset S of an inner product space V is said to be orthogonal if any two distinct vectors in S are orthogonal.

Example:

Consider the set $u = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$, $w = \begin{bmatrix} -5 \\ -2 \\ 1 \end{bmatrix}$

$$\text{Here } \langle u, v \rangle = 1(0) + (-2)(1) + (1)(2)$$

$$= 0 - 2 + 2 = 0$$

$$\langle u, w \rangle = (1)(-5) + (-2)(-2) + (1)(1)$$

$$= -5 + 4 + 1 = 0$$

$$\langle v, w \rangle = 0(-5) + (1)(-2) + (2)(1)$$

$$= 0 - 2 + 2 = 0$$

Here $\{u, v, w\}$ is an orthogonal set.

Orthonormal set:

Let S be a set of vectors in an inner product space V . Then S is said to be orthonormal set, if

$$i) x \in S \quad \|x\| = 1$$

$$\Rightarrow \sqrt{\langle x, x \rangle} = 1$$

$$ii) x, y \in S \text{ \& } x \neq y$$

$$\Rightarrow \langle x, y \rangle = 0$$

Thus an orthonormal set is an orthogonal set with the additional property that each vector is of unit length.

Kronecker delta:

A finite set $S = \{v_1, v_2, \dots, v_n\}$ is orthogonal.

$$\text{if } \langle v_i, v_j \rangle = \delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases} \text{ as known as}$$

Kronecker delta.

Theorem (Gram-Schmidt orthonormalisation Process):

Let V be an inner product space

and let $S = \{u_1, u_2, \dots, u_n\}$ be a linearly independent set in V .

Then $\exists$ an orthonormal set of

vectors $\{v_1, v_2, \dots, v_n\}$ of V such that,
 $\text{span}\{u_1, u_2, \dots, u_n\} = \text{span}\{v_1, v_2, \dots, v_n\}$

Norm (or) length of a vector in an inner product:

Let V be an inner product space. For $x \in V$, we define the norm or length of x , written as $\|x\|$ is defined as the positive square root of $\langle x, x \rangle$

$$(i.e.) \|x\| = \sqrt{\langle x, x \rangle}$$

Example:

Let $V = \mathbb{F}^n$ if $x = (a_1, a_2, \dots, a_n)$

$$\text{then } \|x\| = \sqrt{\langle x, x \rangle} = \left[\sum_{i=1}^n a_i \overline{a_i} \right]^{1/2}$$

[$\because$ By std inner product]

$$= \left[\sum_{i=1}^n |a_i|^2 \right]^{1/2}$$

$= \sqrt{|a_1|^2 + |a_2|^2 + \dots + |a_n|^2}$ is the length of vector.

$x = (a_1, a_2, \dots, a_n)$ in the n -dimensional Euclidean space.

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Working Procedure:
(Gram Schmidt Orthonormalization process)

Let $S = \{u_1, u_2, u_3\}$

step I: $v_1 = \frac{u_1}{\|u_1\|}$

step II: $v_2 = \frac{w_2}{\|w_2\|}$

where $w_2 = u_2 - \langle u_2, v_1 \rangle v_1$

step III: $v_3 = \frac{w_3}{\|w_3\|}$

where $w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$ and so on.

step IV:

Hence, we found the orthonormal basis with respect to $\{u_1, u_2, u_3\}$
(i.e) $\beta = \{v_1, v_2, v_3\}$ is a required orthonormal basis.

step V:

Given α be a vector. Then $\langle \alpha, v_1 \rangle, \langle \alpha, v_2 \rangle$ & $\langle \alpha, v_3 \rangle$ are the required Fourier coefficients.

3)i) Apply Gram-Schmidt process to the subset $S = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$ of $\mathbb{R}^3$ to obtain an orthonormal basis. Also find Fourier coefficients to the basis orthonormal basis of $x = (1, 0, 1)$.

Given $S = \{(1, 1, 1), (0, 1, 1), (0, 0, 1)\}$

Let $u_1 = (1, 1, 1)$; $u_2 = (0, 1, 1)$; $u_3 = (0, 0, 1)$

step I: set $v_1 = \frac{u_1}{\|u_1\|}$

where $\|u_1\| = \sqrt{\langle u_1, u_1 \rangle} = \sqrt{(1, 1, 1) \cdot (1, 1, 1)}$
 $= \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$

$\therefore v_1 = \frac{1}{\sqrt{3}}(1, 1, 1)$

$v_1 = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$ ①

step II: set $v_2 = \frac{w_2}{\|w_2\|}$

where $w_2 = u_2 - \langle u_2, v_1 \rangle v_1$

$\langle u_2, v_1 \rangle = (0, 1, 1) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$

$= 0 + \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}}$

Now, $w_2 = (0, 1, 1) - \frac{2}{\sqrt{3}} \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$ [By ①]

$= (0, 1, 1) - \left(\frac{2}{3}, \frac{2}{3}, \frac{2}{3} \right)$

$$= \left(-\frac{2}{3}, 1-\frac{2}{3}, 1-\frac{2}{3}\right)$$

$$= \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$\text{where } \|w_2\| = \sqrt{\langle w_2, w_2 \rangle} = \sqrt{\left(-\frac{2}{3}\right)^2 + \left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)^2}$$

$$= \sqrt{\frac{4}{9} + \frac{1}{9} + \frac{1}{9}} = \sqrt{\frac{6}{9}} = \frac{\sqrt{2}}{\sqrt{3}}$$

$$\therefore v_2 = \frac{w_2}{\|w_2\|} = \frac{1}{\frac{\sqrt{2}}{\sqrt{3}}} \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$= \frac{\sqrt{3}}{\sqrt{2}} \left(-\frac{2}{3}, \frac{1}{3}, \frac{1}{3}\right)$$

$$v_2 = \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right) \rightarrow \textcircled{2}$$

step III:

$$\text{set } v_3 = \frac{w_3}{\|w_3\|}$$

$$\text{where } w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$$

$$\langle u_3, v_1 \rangle = (0, 0, 1) \cdot \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$= 0 + 0 + \frac{1}{\sqrt{3}} = \frac{1}{\sqrt{3}}$$

$$\langle u_3, v_2 \rangle = (0, 0, 1) \cdot \left(-\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$= 0 + 0 + \frac{1}{\sqrt{6}} = \frac{1}{\sqrt{6}}$$

$$\text{Now, } w_3 = (0, 0, 1) - \frac{1}{\sqrt{3}} \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) - \frac{1}{\sqrt{6}}$$

$$\left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$= (0, 0, 1) - \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right) = \left(-\frac{2}{6}, \frac{1}{6}, \frac{1}{6}\right)$$

$$= \left(0 - \frac{1}{3} + \frac{1}{3}, 0 - \frac{1}{3} - \frac{1}{6}, 1 - \frac{1}{3} - \frac{1}{6}\right)$$

$$= \left(0, -\frac{3}{6}, \frac{2}{6}\right)$$

$$w_3 = \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

$$\text{Also, } \|w_3\| = \sqrt{0^2 + \left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}$$

$$= \sqrt{0 + \frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{2}{4}}$$

$$= \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{2}{\sqrt{2}}\right) = e_3$$

$$\therefore v_3 = \frac{1}{\sqrt{2}} \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

$$= \frac{1}{\sqrt{2}} \left(0, -\frac{1}{2}, \frac{1}{2}\right)$$

$$v_3 = \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \rightarrow \textcircled{3}$$

step IV:

From ①, ② & ③, we get the orthonormal basis:

$$\beta = \{v_1, v_2, v_3\}$$

$$= \left\{ \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right), \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) \right\}$$

Verification:

$$i) \|v_1\| = \sqrt{\frac{1}{3} + \frac{1}{3} + \frac{1}{3}} = \sqrt{\frac{3}{3}} = 1$$

$$\|v_2\| = \sqrt{\frac{4}{6} + \frac{1}{6} + \frac{1}{6}} = \sqrt{\frac{6}{6}} = 1$$

$$\|v_3\| = \sqrt{0 + \frac{1}{2} + \frac{1}{2}} = \sqrt{\frac{2}{2}} = 1$$

$$ii) \langle v_1, v_2 \rangle = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$= \frac{-2}{\sqrt{18}} + \frac{1}{\sqrt{18}} + \frac{1}{\sqrt{18}}$$

$$= \frac{-2}{\sqrt{18}} + \frac{2}{\sqrt{18}} = 0$$

$$\langle v_2, v_3 \rangle = \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right) \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$$= 0 - \frac{1}{\sqrt{12}} + \frac{1}{\sqrt{12}} = 0$$

$$\langle v_1, v_3 \rangle = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right) \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$$= 0 - \frac{1}{\sqrt{6}} + \frac{1}{\sqrt{6}} = 0$$

To find the Fourier coefficient of

$$x = (1, 0, 1):$$

$$\langle x, v_1 \rangle = (1, 0, 1) \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$$

$$= \frac{1}{\sqrt{3}} + 0 + \frac{1}{\sqrt{3}} = \frac{2}{\sqrt{3}} \rightarrow \textcircled{4}$$

$$\langle x, v_2 \rangle = (1, 0, 1) \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$$

$$= \frac{-2}{\sqrt{6}} + 0 + \frac{1}{\sqrt{6}} = \frac{-1}{\sqrt{6}} \rightarrow \textcircled{5}$$

$$\langle x, v_3 \rangle = (1, 0, 1) \cdot \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

$$= 0 + 0 + \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} \rightarrow \textcircled{D}$$

Thus, $\frac{2}{\sqrt{3}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{2}}$ are the required Fourier coefficient. [By ④, ⑤ & ⑥].

Verification:

$$x = \langle x, v_1 \rangle v_1 + \langle x, v_2 \rangle v_2 + \langle x, v_3 \rangle v_3$$

$$= \frac{2}{\sqrt{3}} \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) + \frac{1}{\sqrt{6}} \left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \right)$$

$$+ \frac{1}{\sqrt{2}} \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) = \langle e_1, e_1 \rangle$$

$$= (1, 0, 1) //$$

3) ii) Apply Gram-Schmidt process to the subset $S = \{(1, 0, 1), (0, 1, 1), (1, 3, 3)\}$ of $\mathbb{R}^3$ to obtain an orthonormal basis. Also find Fourier coefficient of $(1, 1, 2)$ relative to the basis orthonormal basis.

Given that $V = \mathbb{R}^3$

Let $x = (a_1, a_2, a_3), y = (b_1, b_2, b_3)$

$a_i, b_i \in \mathbb{R}$. Inner product is defined $\mathbb{R}^3$

$$\langle x, y \rangle = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Here $S = \{(1, 0, 1), (0, 1, 1), (1, 3, 3)\}$

Let $u_1 = (1, 0, 1); u_2 = (0, 1, 1); u_3 = (1, 3, 3)$

Step I:
$$v_1 = \frac{u_1}{\|u_1\|}$$

where $\|u_1\|^2 = \langle u_1, u_1 \rangle$

$$= 1^2 + 0^2 + 1^2$$

$$= 2$$

$$\|u_1\| = \sqrt{2}$$

$$v_1 = \frac{1}{\sqrt{2}}(1, 0, 1)$$

$$v_1 = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \rightarrow \textcircled{1}$$

Step II:

$$w_2 = u_2 - \langle u_2, v_1 \rangle v_1$$

Now, $\langle u_2, v_1 \rangle = \langle (0, 1, 1), \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) \rangle$

$$= 0 + 0 + \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$$

Now, $w_2 = (0, 1, 1) - \frac{1}{\sqrt{2}} \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$

$$= \left(-\frac{1}{2}, 1, \frac{1}{2}\right)$$

$$v_2 = \frac{w_2}{\|w_2\|}$$

where $\|w_2\|^2 = \langle w_2, w_2 \rangle$

$$= \left(-\frac{1}{2}\right)^2 + (1)^2 + \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{4} + 1 + \frac{1}{4} = \frac{1+1+1}{4}$$

$$= \frac{3}{4}$$

$$\|w_2\| = \sqrt{\frac{3}{4}}$$

$$v_2 = \frac{\sqrt{2}}{\sqrt{3}} \left(-\frac{1}{2}, 1, \frac{1}{2}\right)$$

$$v_2 = \left(-\frac{1}{\sqrt{6}}, \frac{\sqrt{2}}{\sqrt{3}}, \frac{1}{\sqrt{6}}\right) \rightarrow \textcircled{2}$$

Step III

$$w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$$

$$\text{Now, } \langle u_3, v_1 \rangle = (1, 3, 3) \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right) \\ = \frac{1}{\sqrt{2}} + 0 + \frac{3}{\sqrt{2}} = \frac{4}{\sqrt{2}}$$

$$\langle u_3, v_1 \rangle = 2\sqrt{2}$$

$$\langle u_3, v_2 \rangle = (1, 3, 3) \left(\frac{-1}{\sqrt{6}}, \frac{\sqrt{2}}{\sqrt{3}}, \frac{1}{\sqrt{6}} \right) \\ = \frac{-1}{\sqrt{6}} + \sqrt{6} + \frac{3}{\sqrt{6}} \\ = \frac{-1+6+3}{\sqrt{6}} = \frac{8}{\sqrt{6}}$$

$$\langle u_3, v_2 \rangle = \frac{8}{\sqrt{6}}$$

$$w_3 = (1, 3, 3) - 2\sqrt{2} \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right) - \frac{8}{\sqrt{6}} \left(\frac{-1}{\sqrt{6}}, \frac{\sqrt{2}}{\sqrt{3}}, \frac{1}{\sqrt{6}} \right) \\ = (1, 3, 3) - (2, 0, 2) + \left(\frac{8}{6}, -\frac{8}{3}, -\frac{8}{6} \right) \\ = \left(1-2+\frac{4}{3}, 3-0-\frac{8}{3}, 3-2-\frac{4}{3} \right) \\ = \left(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3} \right)$$

$$v_3 = \frac{w_3}{\|w_3\|} \quad \text{where } \|w_3\|^2 = \frac{1}{9} + \frac{1}{9} + \frac{1}{9}$$

$$\langle w_3, w_3 \rangle = \frac{3}{9} \\ \left(\frac{1}{3} \right)^2 + \left(\frac{1}{3} \right)^2 + \left(\frac{1}{3} \right)^2 = \frac{1}{3}$$

$$\therefore \|w_3\| = \frac{1}{\sqrt{3}}$$

$$v_3 = \sqrt{3} \left(\frac{1}{3}, \frac{1}{3}, -\frac{1}{3} \right) = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right)$$

$$v_3 = \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right) \rightarrow \textcircled{5}$$

Step IV

From ①, ② & ③ we have found an

orthonormal basis. $\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right), \left(\frac{-1}{\sqrt{6}}, \frac{\sqrt{2}}{\sqrt{3}}, \frac{1}{\sqrt{6}} \right), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right)$

$$\beta = \{v_1, v_2, v_3\}$$

$$= \left\{ \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right), \left(\frac{-1}{\sqrt{6}}, \frac{\sqrt{2}}{\sqrt{3}}, \frac{1}{\sqrt{6}} \right), \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right) \right\}$$

Verification:

$$(1) \|v_1\| = \left(\frac{1}{\sqrt{2}} \right)^2 + 0 + \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{1}{2} + 0 + \frac{1}{2} = \frac{2}{2} = 1$$

$$\|v_2\| = \left(\frac{-1}{\sqrt{6}} \right)^2 + \left(\frac{\sqrt{2}}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{6}} \right)^2$$

$$= \frac{1}{6} + \frac{2}{3} + \frac{1}{6} = \frac{1+4+1}{6} = \frac{6}{6} = 1$$

$$\|v_3\| = \left(\frac{1}{\sqrt{3}} \right)^2 + \left(\frac{1}{\sqrt{3}} \right)^2 + \left(\frac{-1}{\sqrt{3}} \right)^2$$

$$= \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{3}{3} = 1$$

$$(2) \langle v_1, v_2 \rangle = \langle v_2, v_3 \rangle, \langle v_3, v_1 \rangle = 0 \text{ (check)}$$

To find the Fourier coefficient

(a) $x = (1, 1, 2)$ relative to β

we have,

$$\langle x, v_1 \rangle = (1, 1, 2) \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right)$$

$$= \frac{1}{\sqrt{2}} + 0 + \frac{2}{\sqrt{2}} = \frac{3}{\sqrt{2}}$$

$$\boxed{\langle x, v_1 \rangle = \frac{3}{\sqrt{2}}}$$

$$\langle x, v_2 \rangle = (1, 1, 2) \left(\frac{-1}{\sqrt{6}}, \frac{\sqrt{2}}{\sqrt{3}}, \frac{1}{\sqrt{6}} \right)$$

$$= \frac{-1}{\sqrt{6}} + \frac{\sqrt{2}}{\sqrt{3}} + \frac{2}{\sqrt{6}}$$

$$= \frac{-1+2+2}{\sqrt{6}} = \frac{3}{\sqrt{6}}$$

$$\boxed{\langle x, v_2 \rangle = \frac{3}{\sqrt{6}}}$$

$$\langle x, v_3 \rangle = (1, 1, 2) \left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right) = 0$$

$$\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right) \left(\frac{1}{\sqrt{3}} \right) = \frac{1}{\sqrt{3}} + \frac{1}{\sqrt{3}} - \frac{2}{\sqrt{3}} = 0$$

$\therefore \frac{3}{\sqrt{2}}, \frac{3}{\sqrt{6}}, 0$ are Fourier coefficient of x

Verification:

$$x = \langle x, v_1 \rangle v_1 + \langle x, v_2 \rangle v_2 + \langle x, v_3 \rangle v_3$$

$$= (1, 1, 2)$$

15/4/25 Frobenius Inner Product:

Let A, B be any 2×2 matrix

$$\text{Then } \langle A, B \rangle = \text{tr}(A^T B) \quad \langle e_1, e_2 \rangle = \langle e_2, e_1 \rangle = 0$$

3) iii) Apply Gram-Schmidt process to the subset

$$S = \left\{ \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 9 \\ 5 & -1 \end{bmatrix}, \begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix} \right\} \text{ of } M_{2 \times 2}(\mathbb{R})$$

to obtain orthonormal basis using the Frobenius

inner product. Also find the Fourier

coefficients of $\begin{bmatrix} -1 & 27 \\ -4 & 8 \end{bmatrix}$ relative to the orthonormal basis.

$$S = \left\{ \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 9 \\ 5 & -1 \end{bmatrix}, \begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix} \right\}$$

$$\text{Let } u_1 = \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}, u_2 = \begin{bmatrix} -1 & 9 \\ 5 & -1 \end{bmatrix}, u_3 = \begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix} \in M_{2 \times 2}(\mathbb{R})$$

Frobenius Inner Product

$$\langle u_1, u_2 \rangle = \text{tr}(u_1^T u_2)$$

step I: $v_1 = \frac{u_1}{\|u_1\|}$

where, $\|u_1\|^2 = \langle u_1, u_1 \rangle$

$$= \text{tr}(u_1^T u_1)$$

$$= \text{tr} \left[\begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}^T \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} \right]$$

$$= \text{tr} \left[\begin{bmatrix} 3 & -1 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} \right]$$

$$= \text{tr} \begin{bmatrix} 9+1 & 15-1 \\ 15-1 & 25+1 \end{bmatrix}$$

$$= \text{tr} \begin{bmatrix} 10 & 14 \\ 14 & 26 \end{bmatrix}$$

$$= 10 + 26 = 36$$

$$\Rightarrow \|u_1\|^2 = 36$$

$$\Rightarrow \|u_1\| = \sqrt{36} = 6$$

$$v_1 = \frac{u_1}{\|u_1\|} = \frac{\begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}}{6}$$

$$v_1 = \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} \rightarrow \textcircled{1}$$

step II:

$$v_2 = \frac{u_2}{\|u_2\|}$$

where $u_2 = u_2 - \langle u_2, v_1 \rangle v_1$

$$\langle u_2, v_1 \rangle = \text{tr} [u_2^T v_1]$$

$$= \text{tr} \left(\begin{bmatrix} -1 & 9 \\ 5 & -1 \end{bmatrix}^T \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} \right)$$

$$= \frac{1}{6} \text{tr} \left(\begin{bmatrix} -1 & 5 \\ 9 & -1 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} \right)$$

$$= \frac{1}{6} \text{tr} \begin{bmatrix} -3-5 & -5+5 \\ 27+1 & 45-1 \end{bmatrix}$$

$$= \frac{1}{6} \text{tr} \begin{bmatrix} -8 & 0 \\ 28 & 44 \end{bmatrix}$$

$$= \frac{1}{6} (-8 + 44)$$

$$= \frac{1}{6} (36)$$

$$\langle u_2, v_1 \rangle = 6$$

$$\text{Now, } w_2 = u_2 - \langle u_2, v_1 \rangle v_1$$

$$= \begin{bmatrix} -1 & 9 \\ 5 & -1 \end{bmatrix} - 6 \times \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} \quad [\text{By } \textcircled{1}]$$

$$= \begin{bmatrix} -1 & 9 \\ 5 & -1 \end{bmatrix} - \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix}$$

$$w_2 = \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix}$$

$$\|w_2\|^2 = \frac{\langle w_2, w_2 \rangle}{\langle v, v \rangle} = \frac{w_2}{\|v\|} = v$$

$$= \text{tr} [w_2^T w_2]$$

$$= \text{tr} \left(\begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} \begin{bmatrix} -4 & 6 \\ 4 & -2 \end{bmatrix} \right) = v$$

$$= \text{tr} \begin{bmatrix} 16+16 & -24-8 \\ -24-8 & 36+4 \end{bmatrix} = v$$

$$= \text{tr} \begin{bmatrix} 32 & -32 \\ -32 & 40 \end{bmatrix}$$

$$= 32 + 40 = 72$$

$$\|w_2\|^2 = 72$$

$$\|w_2\| = \sqrt{72} = \sqrt{36 \times 2}$$

$$\|w_2\| = 6\sqrt{2}$$

$$\therefore V_2 = \frac{w_2}{\|w_2\|} = \frac{\begin{bmatrix} -1 & 1 \\ 6 & -2 \end{bmatrix}}{6\sqrt{2}}$$

$$= \frac{1}{6\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 6 & -2 \end{bmatrix}$$

$$\boxed{V_2 = \frac{1}{6\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 6 & -2 \end{bmatrix}} \rightarrow \textcircled{5}$$

step III:

$$V_3 = \frac{w_3}{\|w_3\|}$$

$$\text{where } w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$$

$$\langle u_3, v_1 \rangle = \text{tr}(u_3^T v_1) = \text{tr}\left(\begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix}^T \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}\right)$$

$$= \text{tr}\left\{\frac{1}{6} \begin{bmatrix} 7 & 2 \\ -17 & -6 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}\right\}$$

$$= \text{tr}\left\{\frac{1}{6} \begin{bmatrix} 21-2 & 35+2 \\ -51+6 & -85-6 \end{bmatrix}\right\}$$

$$= \text{tr}\left\{\frac{1}{6} \begin{bmatrix} 19 & 37 \\ -45 & -91 \end{bmatrix}\right\}$$

$$= \text{tr}\left\{\frac{19}{6} \quad \frac{37}{6} \atop \frac{-45}{6} \quad \frac{-91}{6}\right\}$$

$$= \frac{19}{6} - \frac{91}{6}$$

$$= \frac{-72}{6}$$

$$\boxed{\langle u_3, v_1 \rangle = -12}$$

$$\text{Also } \langle u_3, v_2 \rangle = \text{tr}(u_3^T v_2)$$

$$= \text{tr}\left(\begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix}^T \frac{1}{6\sqrt{2}} \begin{bmatrix} -1 & 1 \\ 6 & -2 \end{bmatrix}\right)$$

$$= \text{tr}\left\{\frac{1}{6\sqrt{2}} \begin{bmatrix} 7 & 2 \\ -17 & -6 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 6 & -2 \end{bmatrix}\right\}$$

$$= \text{tr} \left\{ \frac{1}{6\sqrt{2}} \begin{bmatrix} -28+12 & 28-4 \\ 68-36 & -68+12 \end{bmatrix} \right\}$$

$$= \text{tr} \left\{ \frac{1}{6\sqrt{2}} \begin{bmatrix} -16 & 24 \\ 32 & -56 \end{bmatrix} \right\}$$

$$= \text{tr} \begin{bmatrix} \frac{-16}{6\sqrt{2}} & \frac{24}{6\sqrt{2}} \\ \frac{32}{6\sqrt{2}} & \frac{-56}{6\sqrt{2}} \end{bmatrix} = \text{tr}$$

$$= \frac{-16}{6\sqrt{2}} - \frac{56}{6\sqrt{2}}$$

$$= \frac{-72}{6\sqrt{2}}$$

$$= \frac{-6 \times \sqrt{2} \sqrt{2}}{\sqrt{2}}$$

$$\langle u_3, v_2 \rangle = -6\sqrt{2}$$

$$w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$$

$$= \begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix} - \left(\frac{2+12}{6} \right) \begin{bmatrix} 3 \\ 5 \end{bmatrix} - (-6\sqrt{2}) \frac{1}{6\sqrt{2}}$$

$$\begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix}$$

$$= \begin{bmatrix} 7 & -17 \\ 2 & -6 \end{bmatrix} + \begin{bmatrix} 6 & 10 \\ -2 & 2 \end{bmatrix} + \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix}$$

$$w_2 = \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix}$$

$$\text{Now, } v_3 = \frac{w_3}{\|w_3\|}$$

$$\text{where } \|w_3\|^2 = \langle w_3, w_3 \rangle =$$

$$= \text{tr}(w_3^T w_3)$$

$$= \text{tr} \left[\begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix}^T \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix} \right]$$

$$= \text{tr} \left[\begin{bmatrix} 9 & 6 \\ -3 & -6 \end{bmatrix} \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix} \right]$$

$$= \text{tr} \begin{bmatrix} 81+36 & -27-36 \\ -27-36 & 9+36 \end{bmatrix}$$

$$= \text{tr} \begin{bmatrix} 117 & -63 \\ -63 & 45 \end{bmatrix}$$

$$= 117 + 45$$

$$\|w_3\|^2 = 162$$

$$\|w_3\| = \sqrt{162} = \sqrt{81 \times 2}$$

$$\boxed{\|w_3\| = 9\sqrt{2}}$$

$$v_3 = \frac{w_3}{\|w_3\|}$$

$$= \frac{\begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix}}{9\sqrt{2}}$$

$$\boxed{v_3 = \frac{1}{9\sqrt{2}} \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix}} \rightarrow \textcircled{3}$$

Hence, From ①, ② & ③, we get the required orthonormal basis.

$$\beta = \{v_1, v_2, v_3\}$$

$$= \left\{ \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix}, \frac{1}{6\sqrt{2}} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix}, \frac{1}{9\sqrt{2}} \begin{bmatrix} 7 & -3 \\ 6 & -6 \end{bmatrix} \right\}.$$

9/25 To find the Fourier coefficient:

$$\text{Given, } A = \begin{bmatrix} -1 & 27 \\ -4 & 8 \end{bmatrix}$$

$$\langle A, v_1 \rangle = \text{tr}(A^T v_1)$$

$$= \text{tr} \left\{ \begin{bmatrix} -1 & -4 \\ 27 & 8 \end{bmatrix} \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} \right\}$$

$$= \text{tr} \left\{ \frac{1}{6} \begin{bmatrix} -3+4 & -5-4 \\ 81-8 & 135+8 \end{bmatrix} \right\}$$

$$= \text{tr} \left\{ \frac{1}{6} \begin{bmatrix} 1 & -9 \\ 73 & 143 \end{bmatrix} \right\}$$

$$= \text{tr} \begin{bmatrix} \frac{1}{6} & \frac{-9}{6} \\ \frac{73}{6} & \frac{143}{6} \end{bmatrix}$$

$$= \frac{1}{6} + \frac{143}{6} = \frac{144}{6} = 24$$

$$\boxed{\langle A, V_1 \rangle = 24}$$

$$\|V_1\| = \sqrt{24}$$

$$\langle A, V_2 \rangle = \text{tr}(A^T V_2)$$

$$= \text{tr} \left\{ \begin{bmatrix} -1 & -4 \\ 27 & 8 \end{bmatrix} \frac{1}{6\sqrt{2}} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} \right\}$$

$$= \text{tr} \left\{ \frac{1}{6\sqrt{2}} \begin{bmatrix} -1 & -4 \\ 27 & 8 \end{bmatrix} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} \right\}$$

$$= \text{tr} \left\{ \frac{1}{6\sqrt{2}} \begin{bmatrix} 4-24 & -4+8 \\ -108+48 & 108-16 \end{bmatrix} \right\}$$

$$= \text{tr} \left\{ \frac{1}{6\sqrt{2}} \begin{bmatrix} -20 & 4 \\ -60 & 92 \end{bmatrix} \right\}$$

$$= \text{tr} \begin{bmatrix} \frac{-20}{6\sqrt{2}} & \frac{4}{6\sqrt{2}} \\ \frac{-60}{6\sqrt{2}} & \frac{92}{6\sqrt{2}} \end{bmatrix}$$

$$= \frac{-20}{6\sqrt{2}} + \frac{92}{6\sqrt{2}} = \frac{72}{6\sqrt{2}}$$

$$= \frac{6 \times \sqrt{2} \sqrt{2}}{\sqrt{2}} = 6\sqrt{2}$$

$$\boxed{\langle A, V_2 \rangle = 6\sqrt{2}}$$

$$\langle V_2, A \rangle = \langle A, V_2 \rangle$$

$$\langle A, v_3 \rangle = \text{tr}(A^T v_3)$$

$$= \text{tr} \left\{ \begin{bmatrix} -1 & -4 \\ 27 & 8 \end{bmatrix} \frac{1}{9\sqrt{2}} \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix} \right\}$$

$$= \text{tr} \left\{ \frac{1}{9\sqrt{2}} \begin{bmatrix} -9 & -24 & 3+24 \\ 243+48 & -81-48 \end{bmatrix} \right\}$$

$$= \text{tr} \left\{ \frac{1}{9\sqrt{2}} \begin{bmatrix} -33 & 27 \\ 291 & -129 \end{bmatrix} \right\}$$

$$= \text{tr} \left\{ \begin{bmatrix} \frac{-33}{9\sqrt{2}} & \frac{27}{9\sqrt{2}} \\ \frac{291}{9\sqrt{2}} & \frac{-129}{9\sqrt{2}} \end{bmatrix} \right\}$$

$$= \frac{-33}{9\sqrt{2}} - \frac{129}{9\sqrt{2}} = \frac{-162}{9\sqrt{2}}$$

$$= \frac{-9\sqrt{2}\sqrt{2}}{\sqrt{2}}$$

$$\boxed{\langle A, v_3 \rangle = -9\sqrt{2}}$$

Thus,

$\{24, 6\sqrt{2}, -9\sqrt{2}\}$ are Fourier coefficients of A relative to β .

Verification:

$$A = \langle A, v_1 \rangle v_1 + \langle A, v_2 \rangle v_2 + \langle A, v_3 \rangle v_3$$

$$= 24 \times \frac{1}{6} \begin{bmatrix} 3 & 5 \\ -1 & 1 \end{bmatrix} + 6\sqrt{2} \frac{1}{6\sqrt{2}} \begin{bmatrix} -4 & 4 \\ 6 & -2 \end{bmatrix} +$$

$$(-9\sqrt{2}) \frac{1}{9\sqrt{2}} \begin{bmatrix} 9 & -3 \\ 6 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 27 \\ -4 & 8 \end{bmatrix} //$$

$$\begin{bmatrix} 12-4-9 & 20+4+3 \\ -4+6-6 & 4-2+6 \end{bmatrix} = \begin{bmatrix} -1 & 27 \\ -4 & 8 \end{bmatrix}$$

3) iv) Apply Gram-Schmidt process to obtain an orthonormal basis for $V = P_2(\mathbb{R})$ with inner product defined by $\langle f, g \rangle = \int_0^1 f(x)g(x)dx$. Also find Fourier coefficient of $h(x) = 1+x$ relative to the orthonormal basis.

Here $S = \{1, x, x^2\}$
 $u_1 = 1; u_2 = x, u_3 = x^2$

Step-1: $v_1 = \frac{u_1}{\|u_1\|} = \frac{1}{1} = 1$

$v_1 = 1 \rightarrow \textcircled{1}$

Step-2:

$v_2 = \frac{w_2}{\|w_2\|}$

$w_2 = u_2 - \langle u_2, v_1 \rangle v_1$

Now, $\langle u_2, v_1 \rangle = \langle x, 1 \rangle$

$= \int_0^1 (x \cdot 1) dx$

$= \left[\frac{x^2}{2} \right]_0^1 = \frac{1}{2}$

$w_2 = x - \frac{1}{2}(1)$

$w_2 = x - \frac{1}{2}$

$\|w_2\|^2 = \langle w_2, w_2 \rangle$

$$= \int_0^1 (x - \frac{1}{2})(x - \frac{1}{2}) dx$$

$$= \int_0^1 (x - \frac{1}{2})^2 dx$$

$$= \left[\frac{(x - \frac{1}{2})^3}{3} \right]_0^1$$

$$= \frac{(\frac{1}{2})^3}{3} - \frac{(-\frac{1}{2})^3}{3}$$

$$= \frac{1}{24} + \frac{1}{24} = \frac{2}{24} = \frac{1}{12}$$

$$\|w_2\| = \frac{1}{\sqrt{12}}$$

$$\therefore v_2 = \frac{1}{(\frac{1}{\sqrt{12}})} (x - \frac{1}{2})$$

$$\text{since } \|w_2\| = \frac{1}{\sqrt{12}} = \frac{1}{2\sqrt{3}}$$

$$\therefore v_2 = 2\sqrt{3}(x - \frac{1}{2}) \rightarrow \textcircled{2}$$

step 3:

$$v_3 = \frac{w_3}{\|w_3\|}$$

$$w_3 = u_3 - \langle u_3, v_1 \rangle v_1 - \langle u_3, v_2 \rangle v_2$$

For,

$$\langle u_3, v_1 \rangle = \langle x^2, 1 \rangle$$

$$= \int_0^1 x^2 \cdot 1 dx$$

$$= \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3}$$

$$\langle u_3, v_1 \rangle = \frac{1}{3}$$

$$\langle u_3, v_2 \rangle = \langle x^2, 2\sqrt{3}(x - \frac{1}{2}) \rangle$$

$$= \int_0^1 x^2 \cdot 2\sqrt{3}(x - \frac{1}{2}) dx$$

$$= 2\sqrt{3} \int_0^1 (x^3 - \frac{1}{2}x^2) dx$$

$$= 2\sqrt{3} \left[\frac{x^4}{4} - \frac{1}{2} \frac{x^3}{3} \right]_0^1$$

$$= 2\sqrt{3} \left[\left(\frac{1}{4} - \frac{1}{6} \right) - (0) \right]$$

$$= 2\sqrt{3} \left[\frac{6-4}{24} \right]$$

$$= 2\sqrt{3} \left[\frac{2}{24} \right]$$

$$= \frac{\sqrt{3}}{6} = \frac{\sqrt{3}}{2 \times 3}$$

$$= \frac{1}{2\sqrt{3}}$$

Now,

$$w_3 = x^2 - \frac{1}{3}(1) - \frac{1}{2\sqrt{3}} 2\sqrt{3}(x - \frac{1}{2})$$

$$= x^2 - \frac{1}{3} - x + \frac{1}{2}$$

$$w_3 = x^2 - x + \frac{1}{6}$$

$$\|w_3\| = \langle w_3, w_3 \rangle$$

$$= \int_0^1 (x^2 - x + \frac{1}{6})(x^2 - x + \frac{1}{6}) dx$$

$$= \int_0^1 (x^2 - x + \frac{1}{6})^2 dx$$

$$\begin{aligned}
 &= \int_0^1 \left[x^4 + x^2 + \frac{1}{36} - 2x^3 + \frac{1}{3}x^2 - \frac{1}{3}x \right] dx \\
 &= \left[\frac{x^5}{5} + \frac{x^3}{3} + \frac{1}{36}x - \frac{2x^4}{4} + \frac{1}{3} \cdot \frac{x^3}{3} - \frac{1}{3} \cdot \frac{x^2}{2} \right]_0^1 \\
 &= \frac{1}{5} + \frac{1}{3} + \frac{1}{36} - \frac{2}{4} + \frac{1}{9} - \frac{1}{6} \\
 &= \frac{1}{5} + \left(\frac{12+1-18+4-6}{36} \right) = \frac{x^5}{5} + \frac{x^3}{3} + \frac{x}{36} \\
 &= \frac{1}{5} + \left(\frac{17-24}{36} \right) = \frac{2x^4}{4} + \frac{x^3}{3 \times 3} - \frac{x^2}{3 \times 2} \\
 &= \frac{1}{5} - \frac{7}{36} =
 \end{aligned}$$

$$= \frac{36-35}{5 \times 36} \Rightarrow \|w_3\|^2 = \frac{1}{5 \times 36}$$

$$\|w_3\| = \frac{1}{6\sqrt{5}}$$

$$v_3 = 6\sqrt{5} \left(x^2 - x + \frac{1}{6} \right) \rightarrow \textcircled{3}$$

From ①, ② & ③, we have found an orthonormal basis:

$$\begin{aligned}
 \beta &= \left\{ v_1, v_2, v_3 \right\} \\
 &= \left\{ 1, 2\sqrt{3} \left(x - \frac{1}{2} \right), 6\sqrt{5} \left(x^2 - x + \frac{1}{6} \right) \right\}
 \end{aligned}$$

To Find Fourier coefficient:

for $h(x) = 1+x$ relative to β

$$\begin{aligned}
 \langle h(x), v_1 \rangle &= \int_0^1 (1+x) \cdot 1 \, dx \\
 &= \left(x + \frac{x^2}{2} \right)_0^1 = 1 + \frac{1}{2} = \frac{3}{2}
 \end{aligned}$$

$$\langle h(x), v_2 \rangle = \int_0^1 (1+x) 2\sqrt{3} \left(x - \frac{1}{2} \right) dx$$

$$\begin{aligned}
 \int_0^1 \left(x - \frac{1}{2} + x^2 - \frac{x}{2} \right) dx &= 2\sqrt{3} \int_0^1 \left(x^2 + \frac{x}{2} - \frac{1}{2} \right) dx \\
 &= 2\sqrt{3} \left[\frac{x^3}{3} + \frac{x^2}{4} - \frac{1}{2}x \right]_0^1 \\
 &= 2\sqrt{3} \left[\frac{1}{3} + \frac{1}{4} - \frac{1}{2} \right] \\
 &= 2\sqrt{3} \left[\frac{4+3-6}{12} \right] = 2\sqrt{3} \left[\frac{1}{12} \right] \\
 &= \frac{\sqrt{3}}{6} = \frac{\sqrt{3}}{2 \times 3} = \frac{1}{2\sqrt{3}}
 \end{aligned}$$

$$\langle h(x), v_3 \rangle = \int_0^1 (1+x) 6\sqrt{5} \left(x^2 - x + \frac{1}{6} \right) dx$$

$$= 6\sqrt{5} \int_0^1 \left(x^2 - x + \frac{1}{6} + x^3 - x^2 + \frac{x}{6} \right) dx$$

$$\begin{aligned}
 &= 6\sqrt{5} \int_0^1 \left(x^3 - \frac{5}{6}x + \frac{1}{6} \right) dx \\
 &= 6\sqrt{5} \left[\left(\frac{1}{4} - \frac{5}{12} + \frac{1}{6} \right) - (0) \right] \\
 &= 6\sqrt{5} \left[\frac{3-5+2}{12} \right] = \frac{\sqrt{5}}{2} [0] \\
 &= 0
 \end{aligned}$$

Thus, $\{ \frac{3}{2}, \frac{1}{2}\sqrt{3}, 0 \}$ are Fourier coefficients of $h(x) = 1+x$ relative to β

Verification: $\langle h(x), v_1 \rangle v_1 + \langle h(x), v_2 \rangle v_2 + \langle h(x), v_3 \rangle v_3$

$$\begin{aligned}
 h(x) &= \langle h(x), v_1 \rangle v_1 + \langle h(x), v_2 \rangle v_2 + \langle h(x), v_3 \rangle v_3 \\
 &= x + 1
 \end{aligned}$$

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Some Important Inequalities:

- (1) Cauchy-Schwarz Inequality.
- (2) Triangle Inequality
- (3) Polar Inequality
- (4) Parallelogram Law.

Triangle Inequality:

If x, y are vectors in an inner product space then $\|x+y\| \leq \|x\| + \|y\|$.

Proof: consider,

$$\begin{aligned}\|x+y\|^2 &= \langle x+y, x+y \rangle \\ &= \langle x, x+y \rangle + \langle y, x+y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle\end{aligned}$$

$$= \|x\|^2 + \langle x, y \rangle + \overline{\langle x, y \rangle} + \|y\|^2$$

$$= \|x\|^2 + 2 \operatorname{Re} \{ \langle x, y \rangle \} + \|y\|^2$$

$$\leq \|x\|^2 + 2 |\langle x, y \rangle| + \|y\|^2$$

$$\leq \|x\|^2 + 2 \|x\| \|y\| + \|y\|^2$$

[$\because$ by Cauchy-Schwarz Inequality]

$$\Rightarrow \|x+y\|^2 \leq (\|x\| + \|y\|)^2$$

Taking square root on both side we get,

$$\|x+y\| \leq \|x\| + \|y\|$$

Hence proved.

Cauchy-Schwarz Inequality:

In an inner product space $V(F)$, then $|\langle x, y \rangle| \leq \|x\| \|y\|$ for $x, y \in V$

Proof:

case (i):

Suppose $x=0$, then $\|x\|=0$

$$\therefore \|x\| \|y\| = 0 \rightarrow \textcircled{1}$$

Now consider (L.H.S)

$$|\langle x, y \rangle| = |\langle 0, y \rangle|$$

$$= |0| \quad [\because \text{by property of inner product}]$$

$$= 0 \rightarrow \textcircled{2}$$

From $\textcircled{1}$ & $\textcircled{2}$, we have

$$|\langle x, y \rangle| = \|x\| \|y\|$$

case (ii):

Suppose $x \neq 0$, then $\|x\| > 0$

Thus $\frac{1}{\|x\|^2}$ is a +ve real number

Now define a vector,

$$z = y - \frac{\langle y, x \rangle}{\|x\|^2} x \text{ in } V$$

consider,

$$\langle z, z \rangle = \left\langle y - \frac{\langle y, x \rangle}{\|x\|^2} x, y - \frac{\langle y, x \rangle}{\|x\|^2} x \right\rangle$$

$$\|z\|^2 = \left\langle y - \frac{\langle y, x \rangle}{\|x\|^2} x, y - \frac{\langle y, x \rangle}{\|x\|^2} x \right\rangle$$

[$\because$ by Axiom (a) of I.P.]

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$$= \langle y, y - \frac{\langle y, x \rangle}{\|x\|^2} x \rangle - \frac{\langle y, x \rangle}{\|x\|^2} \langle x, y - \frac{\langle y, x \rangle}{\|x\|^2} x \rangle$$

[$\because$ by Axiom (b) of I.P]

$$= \langle y, y \rangle - \langle y, \frac{\langle y, x \rangle}{\|x\|^2} x \rangle - \frac{\langle y, x \rangle}{\|x\|^2} \left\{ \langle x, y \rangle - \right.$$

$$\left. \langle x, \frac{\langle y, x \rangle}{\|x\|^2} x \rangle \right\} [\because \text{by Property of I.P}]$$

$$= \|y\|^2 - \frac{\langle y, x \rangle}{\|x\|^2} \langle y, x \rangle - \frac{\langle y, x \rangle}{\|x\|^2} \langle \overline{y, x} \rangle$$

$$+ \frac{\langle y, x \rangle}{\|x\|^2} \frac{\langle \overline{y, x} \rangle}{\|x\|^2} \langle x, x \rangle$$

$$\|z\|^2 = \|y\|^2 - \frac{|\langle y, x \rangle|^2}{\|x\|^2} - \frac{|\langle y, x \rangle|^2}{\|x\|^2} + \frac{|\langle y, x \rangle|^2}{\|x\|^2}$$

$$\frac{1}{\|x\|^2} \|x\|^2 \langle x, x \rangle =$$

[0.4]

$$\text{Since } \|z\| \geq 0$$

$$\|y\|^2 - \frac{|\langle y, x \rangle|^2}{\|x\|^2} \geq 0$$

$$\|y\|^2 \geq \frac{|\langle y, x \rangle|^2}{\|x\|^2}$$

$$\Rightarrow |\langle y, x \rangle|^2 \leq \|x\|^2 \|y\|^2$$

$$\Rightarrow |\langle x, y \rangle|^2 \leq \|x\|^2 \|y\|^2$$

Squaring on both side

$$\Rightarrow |\langle x, y \rangle| \leq \|x\| \|y\|$$

Hence proved

Polar Identities:

Let V be an inner product space over F . Prove that for all $x, y \in V$

$$a) \langle x, y \rangle = \frac{1}{4} [\|x+y\|^2 - \|x-y\|^2] \text{ if } F = \mathbb{R}$$

$$b) \langle x, y \rangle = \frac{1}{4} \sum_{k=1}^4 i^k \|x + i^k y\|^2$$

$$= \frac{1}{4} [i \|x + y\|^2 - \|x - y\|^2 - i \|x - iy\|^2 + \|x + y\|^2] \text{ if } F = \mathbb{C} (i^2 = -1)$$

Proof:

$$a) \text{ Given that } F = \mathbb{R}, \Rightarrow \langle x, y \rangle = \langle y, x \rangle \rightarrow \textcircled{1}$$

$$\text{consider, } \|x+y\|^2 = \langle x+y, x+y \rangle$$

$$= \langle x, x+y \rangle + \langle y, x+y \rangle$$

$$= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle$$

$$= \|x\|^2 + \langle x, y \rangle + \langle y, x \rangle + \|y\|^2 \quad [\because \text{by } \textcircled{1}]$$

$$= \|x\|^2 + 2\langle x, y \rangle + \|y\|^2 \rightarrow \textcircled{2}$$

$$\text{Also, } \|x-y\|^2 = \langle x-y, x-y \rangle$$

$$= \langle x, x-y \rangle - \langle y, x-y \rangle$$

$$= \langle x, x \rangle - \langle x, y \rangle - \{ \langle y, x \rangle - \langle y, y \rangle \}$$

$$= \|x\|^2 - \langle x, y \rangle - \langle x, y \rangle + \|y\|^2 \quad [\because \text{by } \textcircled{1}]$$

$$= \|x\|^2 - 2\langle x, y \rangle + \|y\|^2 \rightarrow \textcircled{3}$$

$$\textcircled{2} - \textcircled{3} \Rightarrow$$

$$\|x+y\|^2 - \|x-y\|^2 = 4\langle x, y \rangle$$

$$\Rightarrow \langle x, y \rangle = \frac{1}{4} \{ \|x+y\|^2 - \|x-y\|^2 \}$$

Hence (a) proved.

b) we need to prove:

$$\|x+y\|^2 - \|x-y\|^2 + i\|x+iy\|^2 - i\|x-iy\|^2 = 4\langle x, y \rangle$$

Consider,

$$\|x-y\|^2 = \langle x-y, x-y \rangle$$

$$= \langle x, x-y \rangle - \langle y, x-y \rangle$$

$$= \langle x, x \rangle - \langle x, y \rangle - \{ \langle y, x \rangle - \langle y, y \rangle \}$$

$$\Rightarrow \|x\|^2 - \langle x, y \rangle - \langle x, y \rangle + \|y\|^2 \rightarrow (1)$$

$$\|x+y\|^2 = \|x\|^2 + 2 \operatorname{Re} \{ \langle x, y \rangle \} + \|y\|^2 \rightarrow (2)$$

$$(2) - (1) \Rightarrow$$

$$\|x+iy\|^2 - \|x-y\|^2 = 4 \operatorname{Im} \{ \langle x, y \rangle \} \rightarrow (3)$$

Now consider,

$$\|x+iy\|^2 = \langle x+iy, x+iy \rangle$$

$$= \langle x, x+iy \rangle + \langle iy, x+iy \rangle$$

$$= \langle x, x \rangle + \langle x, iy \rangle + \langle iy, x \rangle + \langle iy, iy \rangle$$

$$= \|x\|^2 + i \langle x, y \rangle + i \overline{\langle x, y \rangle} + i \cdot i \langle y, y \rangle$$

$$\|x+iy\|^2 = \|x\|^2 - i \langle x, y \rangle + i \overline{\langle x, y \rangle} - (-1) \|y\|^2$$

$$= \|x\|^2 - i [\langle x, y \rangle - \overline{\langle x, y \rangle}] + \|y\|^2$$

$$= \|x\|^2 - i [2i \operatorname{Im} \{ \langle x, y \rangle \}] + \|y\|^2$$

$$\|x+iy\|^2 = \|x\|^2 + 2 \operatorname{Im} \{ \langle x, y \rangle \} + \|y\|^2$$

And,

$$i\|x+iy\|^2 - i\|x\|^2 + 2i \operatorname{Im} \{ \langle x, y \rangle \} + i\|y\|^2 \rightarrow (4)$$

Also, $\|x-y\|^2 = \|x-iy\|^2 = \langle x-iy, x-iy \rangle$

$$\|x-iy\|^2 = \langle x-iy, x-iy \rangle$$

$$= \langle x, x-iy \rangle + \langle iy, x-iy \rangle$$

$$= \langle x, x \rangle - \langle x, iy \rangle + \langle iy, x \rangle - \langle iy, iy \rangle$$

$$= \|x\|^2 - \langle x, iy \rangle + \langle iy, x \rangle + \|y\|^2$$

$$= \|x\|^2 + i \langle x, y \rangle - i \langle x, y \rangle + \|y\|^2$$

$$= \|x\|^2 + i [2 \operatorname{Im} \langle x, y \rangle] + \|y\|^2$$

$$= \|x\|^2 - 2 \operatorname{Im} \langle x, y \rangle + \|y\|^2$$

And, $\langle x, iy \rangle = i \langle x, y \rangle$ and $\langle iy, x \rangle = -i \langle x, y \rangle$

① $\|x-iy\|^2 = \|x\|^2 - 2 \operatorname{Im} \langle x, y \rangle + \|y\|^2 \rightarrow \textcircled{5}$

② $\|x+iy\|^2 = \|x\|^2 + 2 \operatorname{Im} \langle x, y \rangle + \|y\|^2 \rightarrow \textcircled{6}$

③ Now, $\textcircled{5} + \textcircled{6}$ gives $\|x-y\|^2 + \|x+iy\|^2 = 2\|x\|^2 + 2\|y\|^2$

$$\|x-y\|^2 + \|x+iy\|^2 = 2\|x\|^2 + 2\|y\|^2$$

$$= 2[\|x\|^2 + \|y\|^2]$$

$$= 2\langle x+x, y+y \rangle = 2\langle x, y+y \rangle + 2\langle y, x+x \rangle$$

Hence (b) proved.

Parallelogram Law:

In a inner product space V ,

$$\text{we have } \|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

$$\forall x, y \in V$$

Proof:

From previous problem w.k.t

$$\|x+y\|^2 = \|x\|^2 + 2 \operatorname{Re} \{ \langle x, y \rangle \} + \|y\|^2$$

$$\|x-y\|^2 = \|x\|^2 - 2 \operatorname{Re} \{ \langle x, y \rangle \} + \|y\|^2$$

Then we have,

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2.$$

V. Inner Product Space (IPS)

Summary:

* Introduction to Inner Product Space (IPS)

2m & 8m * IPS - Axioms & Properties

Problems (4)

* Norms, Orthogonal & orthonormal set

8m & 2m * Some Important Identities

→ Cauchy - Schwartz Inequality (1)

→ Triangle - Inequality (1)

→ Polar Identity (1)

→ Parallelogram law (1)

2m & 16m * Gram-Schmidt orthonormalization process

→ 8 tuples (2)

→ Integration (1)

→ Frobenius IPS (Matrix - $M_{2 \times 2}(\mathbb{R})$) (1)